ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** heat is used to change the state	Latent heat idea.
2	**(c)** 100 g	Mass is conserved.
3	**(d)** it changed to vapour and mixed with air	Matter is conserved.
4	**(d)** melting and boiling	States change.
5	**(a)** 10 mL	50/5.
6	**(a)** smaller volume and large surface to depth ratio	Less water.
7	**(c)** the colour of the container	Not relevant.
8	**(c)** 60 L	2 x 30.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the fixed boiling temperature does not explain why evaporation occurs at every temperature.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Wind removes vapour near the cloth and speeds up evaporation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Vapour cannot escape (1); Air in the bag becomes saturated (1).	1+1
12	Air's vapour touches the cold surface (1); It condenses on the outside (1).	1+1
13	Fall (1); Wells may dry (1).	1+1
14	Pressure inside is higher (1); Higher pressure raises the boiling point (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	4 x 30 = 120 L (1); 120/2000 (1); 6 per cent (1).	1+1+1
16	Moist air rises over hills (1); It cools and condenses (1); Clouds drop rain (1).	1+1+1
17	Trees release water vapour into the air (1); Roots hold the soil and help water to seep into the ground (1); Fewer floods and more groundwater (1).	1+1+1
	OR: Cover a bowl and an open bowl (1); Open bowl loses level (1); Covered shows droplets on the lid (1).	1+1+1
18	Temperature is below 0 °C (1); Snow is compact (1); Insulates inside (1).	1+1+1
	OR: Forms a barrier (1); Stops vapour escaping (1); Saves water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Evaporation (1). (ii) Higher temperature (or air movement, or spreading out) (1). (iii) It turns into water vapour in the air (1). (iv) The air is already full of moisture (humid), so less water can evaporate (1), and the temperature is lower in the shade or clouds (1).	1+1+1+2
20	Sea water in a basin (1); Sun heats, evaporation (1); Vapour condenses on the cool cover (1); Drops collect in a container (1); Salt left behind (1).	1+1+1+1+1
	OR: Taps at home (1); Reuse water (1); Industrial recycling (1); Dams and canals (1); Policies and awareness (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing evaporation with **boiling**: evaporation happens at any temperature from the surface, boiling at about 100 degrees C.
- Mixing condensation (vapour to liquid) with evaporation (liquid to vapour).
- Calling the white cloud over a boiling kettle water vapour; vapour is invisible, the white cloud is tiny droplets.
- Saying that drops on a cold glass come from inside the glass; they come from **vapour in the air**.
- Writing that humid air helps drying; more moisture in the air makes evaporation slower.
- Forgetting that ice melts and water freezes at the same temperature, 0 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 180 mL	3600/20.
2	**(c)** it increases	More evaporation.
3	**(c)** evaporation and condensation	Local water cycle.
4	**(d)** 50 L	0.05 m^3 = 50 L.
5	**(c)** transpiration part of the cycle	Water vapour from leaves.
6	**(d)** steam releases heat on condensing	Condensation releases heat.
7	**(d)** salts brought by rivers and left after evaporation	Accumulation.
8	**(d)** more transpiration adds moisture	Water vapour to the air.
9	**(c)** A is true but R is false.	A is true: water vapour from air condenses on the cold surface. R is false, glass has no holes.
10	**(d)** A is false but R is true.	A is false: that is evaporation. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Only vapour rises (1); Salt is left behind (1).	1+1
12	Evaporation balances (1); Water returns to the cycle (1).	1+1
13	Ice occupies more volume (1); Expansion on freezing (1).	1+1
14	Evaporation takes heat (1); Cooling effect (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	86400 drops (1); 86400 / 20 = 4320 mL (1); 4.32 L (1).	1+1+1
16	The Sun heats the water (1); Water evaporates into the air (1); Some water also seeps into the ground (1).	1+1+1
17	Heat (1); Moving air (1); Tumbling increases exposure (1).	1+1+1
	OR: Pressure builds (1); Steam expands (1); Vessel bursts (1).	1+1+1
18	Mulching (1); Planting trees as windbreaks (1); Irrigating in the evening (1).	1+1+1
	OR: Water is removed by drying (1); Preserves food (1); Sun used (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Condensation (1). (ii) Water vapour (1). (iii) Melting (1). (iv) Water vapour present in the air touches the cold surface and cools (1), so it changes into tiny liquid drops (1).	1+1+1+2
20	Uneven rain (1); Overuse of groundwater (1); Harvest rain (1); Efficient irrigation (1); Protect ponds (1).	1+1+1+1+1
	OR: Surface: rivers and lakes (1); Groundwater: wells (1); Surface prone to pollution (1); Groundwater limited (1); Both need care (1).	1+1+1+1+1

Mistakes to avoid:

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- Calling the white cloud over a boiling kettle water vapour; vapour is invisible, the white cloud is tiny droplets.
- Saying that drops on a cold glass come from inside the glass; they come from **vapour in the air**.
- Writing that humid air helps drying; more moisture in the air makes evaporation slower.
- Forgetting that ice melts and water freezes at the same temperature, 0 degrees C.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** building percolation pits	Allows seepage.
2	**(d)** snow melts	Glaciers feed rivers.
3	**(d)** condensation	Cooled vapour.
4	**(d)** less evaporation and rain	Less energy.
5	**(d)** burning of wood	Not water movement.
6	**(a)** evaporation from pores takes away heat	Evaporative cooling.
7	**(a)** increases	Ice is less dense.
8	**(a)** ice occupies more volume than water	Expands on freezing.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Vapour cools high up and condenses into tiny droplets that make clouds.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true: melting and freezing happen at the same temperature, but one does not explain the other.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They store water as ice (1); Melt in summer to feed rivers (1).	1+1
12	Air near the ground cools (1); Vapour condenses into tiny droplets (1).	1+1
13	They affect rivers and people (1); Need safe design (1).	1+1
14	Water on the skin evaporates (1); Takes away heat (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Winds move clouds (1); Clouds cool and rain (1); Rain falls far from the source (1).	1+1+1
16	Milk forms a layer of cream (1); Traps steam (1); Bubbles push it up (1).	1+1+1
17	Flood: water overflows land (1); Shower: short rain (1); Drainage determines the result (1).	1+1+1
	OR: Sea water in shallow pans (1); Sun evaporates water (1); Salt crystals remain (1).	1+1+1
18	Water on skin evaporates (1); Evaporation needs heat (1); Heat is taken from skin (1).	1+1+1
	OR: Fast evaporation (1); Water wasted (1); Leaves may burn (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) The water cycle (1). (ii) Evaporation (1). (iii) Tiny water droplets (and ice crystals) (1). (iv) Rain and snow (or hail) (1); it keeps supplying fresh water for living things (1).	1+1+1+2
20	Ice cube to water (1); Water to steam on a stove (1); Steam to droplets on a lid (1); Water to ice in a freezer (1); Same substance, different states (1).	1+1+1+1+1
	OR: Cylinder with funnel (1); Measure depth daily (1); Record monthly total (1); Compare months (1); Use for harvesting plan (1).	1+1+1+1+1

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